§3 The three principles: SM, EU and FR — part 3 of 5

Reference: subsec:eu, subsec:fr, subsec:principle_failure_relation, subsec:principle_linkage

Source: paper/section3_principles.tex, lines 2360–3118 (§3.6 EU; §3.7 FR; §3.8 the three principles and blockage, well-formed realization and collapse; §3.9 linkage of the three principles and limits on implications), at commit e6e12cd. The Japanese manuscript is authoritative. Part 3 of §3 follows 03_principles_2.md. Part 4 of §3 follows in 03_principles_4.md; part 5 follows in 03_principles_5.md.

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§3.6 EU

Reference: subsec:eu

This subsection defines the time-indexed satisfaction predicate of EU. EU does not ask whether all required environmental factors are satisfied at all times. It asks whether a factor is satisfied in the situation where, if it were unsatisfied, no action would remain that puts the next state in K. This situation can arise whether the current state is inside or outside K: inside K, satisfaction is needed to maintain feasibility; outside K, it is needed to recover into K in one step. The situation is fixed by a per-factor one-step critical domain and a one-step critical required factor set. Both are determined only by the required environment space E and the feasible action correspondence A of §2, and no new assumption is used.

Definition (def:eu_critical_domain): one-step critical domain (classification C). For a required environmental factor $\eta \in E$, $\mathrm{Crit}_\eta^{(1)} \subseteq X$ is the set of $x \in X$ satisfying all of:

- (i) there is $e^0 \in E_{\mathrm{ext}}$ with $\eta(x, e^0) = 0$;
- (ii) for all $e' \in E_{\mathrm{ext}}$, $\eta(x, e') = 0 \Rightarrow A(x, e') = \emptyset$;
- (iii) there is $e \in E_{\mathrm{ext}}$ with $\eta(x, e) = 1$ and $A(x, e) \neq \emptyset$.

Each condition depends only on η and A (def:at) and is meaningful for every $x \in X$. (i) excludes the case where (ii) holds vacuously; (ii) says that in every external environment state where the factor is unsatisfied, no action puts the next state in K; (iii) says that among the states where the factor is satisfied there is one with such an action. For $x \in K$, $A(x, e)$ is the set of actions that maintain feasibility; for $x \notin K$ it is the set of actions that recover into K in one step. $\mathrm{Crit}_\eta^{(1)}$ is a one-step condition and is not identified with the topological boundary of K. The paper does not claim $\mathrm{Crit}_\eta^{(1)} \neq \emptyset$ in general.

Definition (def:eu_fire): one-step critical required factor set (classification C). $\mathrm{Crit}^{(1)} : X \to 2^E$, $\mathrm{Crit}^{(1)}(x) := \{\eta \in E \mid x \in \mathrm{Crit}_\eta^{(1)}\}$. Thus $\eta \in \mathrm{Crit}^{(1)}(x) \iff x \in \mathrm{Crit}_\eta^{(1)}$; $\mathrm{Crit}_\eta^{(1)} \subseteq X$ is a set of states and $\mathrm{Crit}^{(1)}(x) \subseteq E$ a set of factors. For times one writes $\mathrm{Crit}^{(1)}(x_t)$, with no time-subscript abbreviation. $\mathrm{Crit}^{(1)}(x)$ is determined by x alone and does not depend on the realized external environment state. $\eta \in \mathrm{Crit}^{(1)}(x)$ is read "η is a critical required factor at x". Every element of E is a required factor, but only those that are critical for one-step feasibility at the state are critical required factors. The superscript (1) marks a one-step condition (rem: eu_one_step_scope).

Description level (rem:crit_description_level). $\mathrm{Crit}^{(1)}(x)$ is a classification-C quantity determined theoretically by η, f and K, describing that a factor is critical for one-step maintenance or recovery of feasibility independently of whether the ACS recognizes it. Three things are distinguished: (1) the theoretical critical required factors $\mathrm{Crit}^{(1)}(x_t)$; (2) the ACS's internal estimate and evaluation of necessity, formed by sensing, discrimination and evaluation and by standard functional forms such as TIPA (§4); (3) measurement and diagnosis by an external observer. The general structure excludes neither a miss (a theoretically critical factor not evaluated as needed internally) nor a false evaluation (a non-critical factor evaluated as needed). The truth condition of the time-indexed predicate of EU (def:EU) is determined by (1) alone and uses neither the internal evaluation (2) nor the measurement (3). The standard descrip-

tion of (2) is the responsibility of §4; the ACS is not required to hold or compute $\mathrm{Crit}^{(1)}(x_t)$ (`rem:s3_description_levels`). The vocabulary is also separated: (1) uses "critical", "necessary", "satisfied"; (2) uses "estimate internally", "detect", "evaluate", "fire", "operate"; (3) uses "the observer estimates", "observes", "measures", "diagnoses". "Estimate" can occur in (2) and (3), so its subject is stated. "Fire" is reserved for internal operation in (2) and is not used for the theoretical quantity in (1).

Definition (`def:EU`): time-indexed satisfaction predicate of EU. For $t \in \mathbb{T}_{\mathrm{P},i}$, EU_t is true if $\forall \eta \in \mathrm{Crit}^{(1)}(x_t) : \eta(x_t, e_t) = 1$, and false otherwise. For $t \notin \mathbb{T}_{\mathrm{P},i}$, EU_t is not defined.

Scope of the truth of EU (`rem:eu_vacuous`). If $\mathrm{Crit}^{(1)}(x_t) = \emptyset$, the universal quantification of `def:EU` holds vacuously and EU_t is true. The defining condition of the truth of EU_t is only that the critical required factors at time t are all satisfied; it does not include satisfaction of non-critical factors. The truth of EU_t therefore does not express that all required environmental factors are available or that the function of utilizing the environment is sound (`rem:eu_env_utilization`). The domain of EU_t does not presuppose the truth of SM_t; SM and EU are judged separately as distinct constraint axes.

Proposition (`prop:eu_blockage`): unsatisfied critical factor and blockage. (i) For any $t \in \mathbb{N}_0$ and $\eta \in E$: $x_t \in \mathrm{Crit}_\eta^{(1)}$ and $\eta(x_t, e_t) = 0$ imply $A_t = \emptyset$. (ii) If $t \in \mathbb{T}_{\mathrm{P},i}$ and EU_t is false, then $A_t = \emptyset$. *Proof idea:* (i) is condition (ii) of `def:eu_critical_domain` with $e' = e_t$; (ii) picks the unsatisfied critical factor and applies (i).

Proposition (`prop:eu_sm_next`): unsatisfied critical factor and the next state. If the antecedent of (i) or (ii) of `prop:eu_blockage` holds, then $x_{t+1} \notin K$; if moreover $t+1 \in \mathbb{T}_{\mathrm{P},i}$, then SM_{t+1} is false. *Proof idea:* $A_t = \emptyset$ and `prop:sm_one_step_A`(ii), with $u_t \in U$; the second part is `def:SM`.

Maintenance and recovery (`rem:eu_maintenance_recovery`). If $x_t \in K$ and EU_t is false, the state leaves K at the next time; if $x_t \notin K$ and EU_t is false, the state is outside K also at the next time. An unsatisfied critical factor thus blocks one-step maintenance of feasibility inside K and one-step recovery into K outside it. In neither case does the paper state, as a general consequence of falsity of EU_t alone, unrecoverability over several steps, $\mathrm{WF}_{i,t} = 0$ or collapse.

Reach of one-step necessity (`rem:eu_one_step_scope`). $\mathrm{Crit}_\eta^{(1)}$ and $\mathrm{Crit}^{(1)}$ are one-step conditions, so two things are not in $\mathrm{Crit}^{(1)}$ or in the truth condition of EU_t: (1) a factor whose one-time lack still leaves an action putting the next state in K, but whose persistent lack loses feasibility; (2) a factor needed to recover into K from a state outside K over several steps. For persistent non-satisfaction in (1), (CR-1a)(ii) of CR-1 treats it at the level of well-formed realization (`rem:eu_cr1_connection`). The connection of multi-step recoverability and finite-time reachability of K with required environmental factors is not formalized in this paper (§9).

Connection of EU with CR-1 (`rem:eu_cr1_connection`). Falsity of EU_t means that some critical factor is unsatisfied at t and, by `prop:eu_blockage`, accompanies blockage at t. Connection to loss of well-formed realization is through (CR-1a)(ii) (`subsec:environment_space`): for one factor η, if for some $t_0 \geq 1$ $\mathrm{WF}_{i,t_0-1} = 1$ and $\eta(x_{t_0-1}, e_{t_0-1}) = 1$, and $\eta(x_s, e_s) = 0$ for all $s \geq t_0$, then $\mathrm{WF}_{i,s} = 0$ for all s after some finite $T \geq t_0$. The paper does not state, as a general consequence of EU_t being false at some time, that this antecedent holds. When the critical factor differs from time to time, or the non-satisfaction does not persist, the antecedent of (CR-1a)(ii) is not met. The consequence of (CR-1a)(ii) is loss of well-formed realization, not functional termination $\mathrm{FT}_{i,t} = 1$ and not collapse $\mathrm{Col}_{i,t} = 1$.

Proposition (`prop:eu_fire_domain`): critical factors and the factor-wise satisfiable-state domain. For $\eta \in E$ with D_η as in `def:eta_satisfiable_domain`: (i) $\mathrm{Crit}_\eta^{(1)} \subseteq D_\eta$, hence $\eta \in \mathrm{Crit}^{(1)}(x) \Rightarrow x \in D_\eta$; (ii) $x \notin D_\eta \Rightarrow \eta \notin \mathrm{Crit}^{(1)}(x)$. *Proof idea:* (iii) of `def:eu_critical_domain` gives an e with $\eta(x, e) = 1$; (ii) is the contrapositive.

Factors unsatisfiable at the current state and EU (`rem:eu_env_utilization`). $x \notin D_\eta$ says that at the current state no external environment state satisfies η. Since $\eta(x, e)$ depends on the internal state x, changes of internal state relevant to intake, acquisition and connection are expressed within this relation. By `prop:eu_fire_domain`(ii), such η is not a critical required factor, and the time-indexed predicate of EU does not ask for its satisfaction. That $E_{\mathrm{req}}(x) = \emptyset$

can occur at such a state follows from (a) of rem:env_utilization_function in §2. The paper does not state, as a general consequence of $x \notin D_\eta$ alone, blockage, $\mathrm{WF}_{i,t} = 0$, functional termination or collapse (same treatment as (c) of that remark). A blockage at such a state is not counted as (B1) of prop:blockage_dichotomy for reason of η's non-satisfaction, and the paper sets no definition that counts this case separately as an EU-side event. The required-environment utilization function of §2 is a collective name for existing functions cooperating to bring the internal state into D_η, and relates to the principle response item of EU. The paper does not state, as a general consequence, truth of EU_t from that function operating, or stoppage of the function from falsity of EU_t (rem:s3_description_levels).

Connection of CR-1 with realized external interaction (rem:cr1_ei_connection). CR-1 specifies each element of E as a substantive required factor for the persistence of the structural self through realizability of deficiency, inability to persist under deficiency, and joint satisfiability. When the antecedent of a persistent deficiency specified by CR-1 holds on a realized trajectory for some $\eta \in E$, the consequence is evaluated according to CR-1. But $\chi_{\mathrm{EI},t} = 0$ alone does not establish that the persistent-deficiency condition of a particular η holds. This is consistent with $\eta : X \times E_{\mathrm{ext}} \to \{0, 1\}$ depending on the internal state x_t: internal reserves, internal compensation or state-dependent tolerance time may mean that a temporary lack of external action does not immediately amount to persistent deficiency, and the Phase 1 general structure does not exclude this. The paper therefore sets no unconditional proposition "terminal complete isolation certainly loses feasibility". To connect terminal complete isolation with inability to persist, one must separately confirm or assume that the isolation actually connects to the persistent-deficiency condition of required factors in CR-1.

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§3.7 FR

Reference: subsec:fr

This subsection defines the time-indexed satisfaction predicate of FR (local FR). Local FR asks, only in situations where all critical required factors at time t are satisfied, whether an action exists that puts the next state in K: inside K this is the existence of an action maintaining feasibility; outside K, of an action recovering into K in one step. A blockage accompanying unsatisfied critical factors is treated as falsity of EU and is not included in the judgment of local FR. The response diversity is also defined as a diagnostic quantity.

Definition (def:fr_domain): domain of FR. $\mathcal{D}_{\mathrm{FR}} := \{(x, e) \in X \times E_{\mathrm{ext}} \mid \forall \eta \in \mathrm{Crit}^{(1)}(x) : \eta(x, e) = 1\}$. Since $\mathrm{Crit}^{(1)}$ is a map on X (def:eu_fire), the condition is determined at every (x, e). If $\mathrm{Crit}^{(1)}(x) = \emptyset$, then $(x, e) \in \mathcal{D}_{\mathrm{FR}}$ for all e.

Definition (def:FR): time-indexed satisfaction predicate of FR (local FR). For $t \in \mathbb{T}_{\mathrm{P},i}$ with $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$, FR_t is true if $A_t \neq \emptyset$ and false if $A_t = \emptyset$. Also called **local FR**. For $t \notin \mathbb{T}_{\mathrm{P},i}$ or $(x_t, e_t) \notin \mathcal{D}_{\mathrm{FR}}$, FR_t is not defined (undefined is not false).

Limits in the definition of FR (rem:fr_scope).

1. "Feasible" is fixed to this paper's feasibility, i.e. $f(x, u, e) \in K$; it does not mean executability of actions in general, success of an attempt, or degree of success of execution (rem:f_reading).
2. FR as a constraint axis concerns obtaining feasible responses where needed, and the retention, acquisition, recovery and reconstruction of the potential for needed responses. Local FR is its one-time aspect; the long-term aspect is partly treated in subsec:sd_necessity and subsec:s3_tstab.
3. On $\mathcal{D}_{\mathrm{FR}}$, falsity of FR_t coincides with blockage $A_t = \emptyset$, but FR is not a rewording of blockage: blockage is a per-time diagnosis of state, whereas FR is a response-side constraint axis that includes long-term aspects.
4. At a time with $x_t \notin K$, local FR asks whether an action recovering into K in one step exists. Falsity of FR_t then means that return to K in one step is not possible and says nothing about the possibility of recovery through several steps (rem:eu_one_step_scope). The ACS may be executing an action that contributes to recovery at that time, and falsity of FR_t does not mean that no such action exists, that the recovery process is not progressing, or that the response mechanism is impaired (rem:blockage_scope_sec2).

5. Self-development is positioned as one mode of the principle response item of acquiring and extending the potential for needed responses.
6. "Response capability" is used only as a general term; the formal term is "potential for needed responses".

Role of the domain (`rem:fr_domain_role`). $\mathcal{D}_{\mathrm{FR}}$ does not change the value of A_t; it fixes which blockages are counted as times at which the time-indexed predicate of FR is false. A blockage accompanying unsatisfied critical factors is treated as falsity of EU and not included in FR (`prop:blockage_dichotomy`). $\mathcal{D}_{\mathrm{FR}}$ does not include $x \in K$ as a condition, so whether one-step recovery outside K is possible is included in the judgment of FR.

Definition (`def:mu_eff`): response diversity (classification C). Fix $(x, e) \in X \times E_{\mathrm{ext}}$. For each $u \in A(x, e)$, the feasibility-retention indicator $\varphi_u : E_{\mathrm{req}}(x) \to \{0, 1\}$ is $\varphi_u(e') := \mathbf{1}[f(x, u, e') \in K]$, and the equivalence $\sim$ on $A(x, e)$ is $u \sim u' \iff \varphi_u = \varphi_{u'}$ as functions on $E_{\mathrm{req}}(x)$. The **response diversity** is the cardinality of the quotient, $\mu_{\mathrm{eff}}(x, e) := |A(x, e)/\!\sim|$, with $\mu_{\mathrm{eff},t} := \mu_{\mathrm{eff}}(x_t, e_t)$. Finiteness of μ_{eff} is not assumed. It is a diagnostic quantity defined at all (x, e) and is used in the definition of none of the time-indexed satisfaction predicates.

Proposition (`prop:fr_domain_eu`): domain and EU. For $t \in \mathbb{T}_{\mathrm{P},i}$: EU_t is true $\iff (x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$. Hence for $t \in \mathbb{T}_{\mathrm{P},i}$, FR_t being defined is equivalent to EU_t being true. *Proof idea:* both conditions are $\forall \eta \in \mathrm{Crit}^{(1)}(x_t) : \eta(x_t, e_t) = 1$.

Proposition (`prop:fr_one_step`): local FR and membership of the next state in K. For $t \in \mathbb{T}_{\mathrm{P},i}$ with $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$: FR_t true $\iff A_t \neq \emptyset \iff \exists u \in U : f(x_t, u, e_t) \in K \iff \exists u \in U : \Pi(f(x_t, u, e_t)) \geq 0$. *Proof idea:* `def:FR` and `prop:sm_one_step_A`(i).

Proposition (`prop:fr_next_sm`): falsity of local FR and the next state. If $t \in \mathbb{T}_{\mathrm{P},i}$, $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$ and FR_t is false, then $A_t = \emptyset$ and $x_{t+1} \notin K$; if moreover $t + 1 \in \mathbb{T}_{\mathrm{P},i}$, SM_{t+1} is false.

Proposition (`prop:fr_mu_eff`): local FR and response diversity. (i) For any (x, e): $A(x, e) = \emptyset \iff \mu_{\mathrm{eff}}(x, e) = 0$. (ii) For $t \in \mathbb{T}_{\mathrm{P},i}$ with $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$: FR_t true $\iff \mu_{\mathrm{eff},t} \geq 1$; in particular $\mu_{\mathrm{eff},t} \geq 2 \Rightarrow \mathrm{FR}_t$ true.

Response diversity when $E_{\mathrm{req}}(x_t) = \emptyset$ (`rem:mu_eff_empty_req`). If $E_{\mathrm{req}}(x_t) = \emptyset$, every φ_u is the unique function on the empty set, so $A_t \neq \emptyset$ gives $\mu_{\mathrm{eff},t} = 1$. This is a set-theoretic consequence of there being no required-environment-satisfying state to compare, and does not show that the possibility of response has degenerated to one kind. `prop:fr_mu_eff` holds in this case too.

Intrinsic comparison basis (`rem:env_basis_intrinsic`). The comparison basis of φ_u, $\sim$ and $\mu_{\mathrm{eff},t}$ is $E_{\mathrm{req}}(x_t)$ of §2, a set derived uniquely from E and x_t under the fixed E_{ext}, not an evaluation region the analyst chooses. No inclusion in either direction is assumed between $E_{\mathrm{req}}(x_t)$ and the one-step feasible environment correspondence $\mathcal{E}^{(1)}_{\mathrm{viab}}(x_t)$ (`def:one_step_viable_environment`).

Local FR versus long-term aspects (`rem:sd_local_scope`). Local FR is a one-step condition relative to the fixed current state x_t and realized environment state e_t. The paper does not state, as a general consequence of FR_t being true at t, that FR_s is defined and true at a later $s > t$; likewise $\mu_{\mathrm{eff},t} \geq 2$ or $= 1$ is a local quantity and fixes no later value. As stated in `subsubsec:real_world_scope`, the ACSs addressed operate in time-varying environments with finite computational resources, response time and memory, supply and energy constraints, and possible damage or degradation. Hence $x_s = x_t$, $e_s = e_t$, $A_s = A_t$, $E_{\mathrm{req}}(x_s) = E_{\mathrm{req}}(x_t)$ are not assumed to be maintained; A_s, $E_{\mathrm{req}}(x_s)$, FR_s, $\mu_{\mathrm{eff},s}$ are evaluated afresh at each time. Analyzing a given realized sequence in Phase 1 does not assume that the ACS or an external observer can know the future sequence in advance (`rem:three_layers`), and Phase 1 introduces no probability measure on future sequences. For the long-term aspects of FR (retention, acquisition, recovery and reconstruction of the potential for needed responses), Phase 1 gives no independent infinite-time judgment condition. Persistence of local FR within a finite window is described in (T2) of `subsec:s3_tstab`, the conditional connection between persistent degeneration of response diversity and long-term feasibility in `subsec:sd_necessity`, and finite-step robustness by Hor_f in `subsec:viab_core`, each partially. Change of the internal state across time and updating of response strategy are the responsibility of §5. A full theory of the long-term aspects is Phase 1.α (§9).

Improvability of the feasibility margin (`rem:sd_improvement_aux`). The condition $\exists u \in A_t : \Pi(f(x_t, u, e_t)) > \Pi(x_t)$ says that under the current realized environment state an action exists that strictly improves the feasibility margin in one step. It is kept as an auxiliary diagnostic condition but is not a condition of local FR (`def:FR`), so that states with already high margin, or states retaining a feasible response while not strictly increasing Π in one step, are not made false for local FR on that ground alone.

Proposition (`prop:viab_stage_fr`): finite-step robust feasible set and local FR. If $\nu \geq 1$, $x_t \in \mathrm{Viab}_f^{(\nu)}(K)$ and $e_t \in E_{\mathrm{req}}(x_t)$, then $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$ and $A_t \neq \emptyset$; if moreover $t \in \mathbb{T}_{\mathrm{P},i}$, FR_t is true. *Proof idea:* `def:viab_stage` gives a common u sending all $e \in E_{\mathrm{req}}(x_t)$ into $\mathrm{Viab}_f^{(\nu-1)}(K) \subseteq K_{\mathrm{req}} \subseteq K$, so $u \in A_t$; $e_t \in E_{\mathrm{req}}(x_t)$ gives $\eta(x_t, e_t) = 1$ for all $\eta \in E \supseteq \mathrm{Crit}^{(1)}(x_t)$.

Proposition (`prop:selection_responsibility`): responsibility of selection. For any $t \in \mathbb{N}_0$, $u_t \notin A_t \Rightarrow x_{t+1} \notin K$. In particular, even if SM_t, EU_t and FR_t are all true at $t \in \mathbb{T}_{\mathrm{P},i}$, $u_t \notin A_t$ gives $x_{t+1} \notin K$. *Proof idea:* `prop:sm_one_step_A`(ii).

Selection responsibility and constitutive requirements (`rem:selection_responsibility`). The paper places $u_t \in A_t$ neither among constitutive requirements nor among assumptions. If the time-indexed predicates of the three principles are all true, $A_t \neq \emptyset$ (`def:FR`), but their definitions do not include $u_t \in A_t$, and the paper does not require $u_t \in A_t$ from their joint truth. Formation of the effective action is the responsibility of TIPA and PSL in §4, which does not require $\mathrm{Cand}_t = A_t$.

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§3.8 The three principles and blockage, well-formed realization and collapse

`Reference: subsec:principle_failure_relation`

This subsection splits blockage into two according to whether critical required factors are satisfied, and organizes the relation between the time-indexed predicates of the three principles and blockage, well-formed realization and collapse. The split applies both to blockage at times when the current state is inside K (feasibility cannot be maintained in one step) and outside K (return to K in one step is not possible).

Proposition (`prop:blockage_dichotomy`): dichotomy of blockage. For a time t with $A_t = \emptyset$, exactly one of (B1) and (B2) holds:

- **(B1) blockage with unsatisfied critical required factor:** there is $\eta \in \mathrm{Crit}^{(1)}(x_t)$ with $\eta(x_t, e_t) = 0$.
- **(B2) blockage under satisfied critical required factors:** $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$.

Also, at any time, the condition of (B1) implies $A_t = \emptyset$. Further, for $t \in \mathbb{T}_{\mathrm{P},i}$: (i) a time of (B1) blockage is a time at which EU_t is false, and then $(x_t, e_t) \notin \mathcal{D}_{\mathrm{FR}}$, so FR_t is **undefined**; (ii) a time of (B2) blockage is a time at which EU_t is true and FR_t is false; (iii) $A_t = \emptyset$ is equivalent to "EU_t is false" or "EU_t is true and FR_t is false", and these two cannot hold simultaneously. *Proof idea:* the condition of (B2) is the negation of that of (B1) because η takes values in $\{0, 1\}$; (B1) implies $A_t = \emptyset$ by `prop:eu_blockage`(i); the rest uses `def:EU`, `prop:fr_domain_eu` and `def:FR`.

Status of the dichotomy (`rem:blockage_dichotomy`). The proposition is a diagnostic partition following from definitions, not a classification of causes of blockage. The names (B1) and (B2) are fixed without time-indexed predicates and so apply also to blockage at times outside $\mathbb{T}_{\mathrm{P},i}$; the correspondence with the predicates is stated only for $t \in \mathbb{T}_{\mathrm{P},i}$. At a (B1) time FR_t is not false but undefined; undefined is not counted as false (`rem:s3_description_levels`). In (B1) blockage the paper does not claim that non-satisfaction of η is the sole cause. A blockage at $x_t \notin K$ means that return to K in one step is not possible and says nothing about recovery through several steps (`rem:eu_one_step_scope`). A factor that is unsatisfiable by any external environment state at the current state is not a critical required factor, so blockage involving such a factor is not counted as (B1) (`rem:eu_env_utilization`). (B2) is the diagnostic residual of blockages not meeting the condition of (B1) and does not represent a response-side failure. For example, if a one-step blockage arises only because several required factors are jointly

unsatisfied and no factor alone is critical, that blockage belongs to (B2). Likewise, if at $x_t \notin K$ return to K in one step is impossible under every external environment state, no factor satisfies (iii) of `def:eu_critical_domain`, the critical set is empty, and the blockage at that time belongs to (B2). The dichotomy is relative to the fixed decomposition E and the one-step judgment, and does not classify causal contributions of the environment side and the response side.

Figure (`fig:s3_predicate_domain`). The manuscript's figure shows (a) the domains of the time-indexed predicates — $\mathrm{FT}_{i,t} = 1$ gives no predicate (neither true nor false); for $\mathrm{FT}_{i,t} = 0$, i.e. $t \in \mathbb{T}_{\mathrm{P},i}$, SM_t and EU_t are judged separately (no line from SM to EU or FR); FR_t is defined only where EU_t is true ($A_t \neq \emptyset$ true, $A_t = \emptyset$ false), and where EU_t is false FR_t is not defined (not false) and $A_t = \emptyset$ — and (b) the dichotomy (B1)/(B2) of all times with $A_t = \emptyset$, with the reading in terms of predicates shown dashed because it holds only inside $\mathbb{T}_{\mathrm{P},i}$. Arrows are case distinctions by definition, not causal or temporal order. The figure itself is not reproduced in this summary.

Time-indexed predicates and blockage, well-formed realization, collapse (`rem:principles_failure_table`). For $t \in \mathbb{T}_{\mathrm{P},i}$:

1. Falsity of SM_t is not a defining condition of blockage; the paper does not state, as a general consequence of falsity of SM_t alone, blockage $A_t = \emptyset$.
2. EU_t false $\Rightarrow A_t = \emptyset$ (`prop:eu_blockage`).
3. FR_t false $\Rightarrow A_t = \emptyset$, and on $\mathcal{D}_{\mathrm{FR}}$ falsity of FR_t coincides with blockage (`def:FR`).
4. Blockage times split exclusively and exhaustively into times with EU_t false and times with EU_t true and FR_t false (`prop:blockage_dichotomy`).
5. The paper does not state, as a general consequence of falsity of any one predicate alone, $\mathrm{WF}_{i,t} = 0$ or collapse $\mathrm{Col}_{i,s} = 1$ ($s \geq t$).
6. At times $t \in \mathbb{T}_{\mathrm{P},i}$ at which predicates are judged, $\mathrm{FT}_{i,t} = 0$ (`def:principle_time_domain`) and $\mathrm{Col}_{i,t} = 0$ (`lem:principle_time_domain_down`).

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§3.9 Linkage of the three principles and limits on implications

Reference: `subsec:principle_linkage`

This subsection collects the conditional implications shown in §3 among the time-indexed predicates and states which implications the paper does not claim.

Linkage (`rem:principle_linkage`). The three principles do not define one another; they are conceptually distinct constraint axes. Conditional implications under explicit premises or domains are not excluded. Those shown in §3 are:

1. `prop:eu_sm_next`: falsity of EU gives $x_{t+1} \notin K$ at the next time.
2. `prop:fr_next_sm`: falsity of local FR gives $x_{t+1} \notin K$.
3. `prop:viab_stage_fr`: membership in a finite-step robust feasible set (with $e_t \in E_{\mathrm{req}}(x_t)$) gives truth of local FR.
4. `prop:fr_domain_eu`: the domain of FR coincides with truth of EU.
5. `prop:blockage_dichotomy`: correspondence of the blockage dichotomy with the predicates of EU and FR.

Limits on implications (`rem:principle_implication_scope`). For $t \in \mathbb{T}_{\mathrm{P},i}$ the paper does not state, as a general consequence, truth of EU_t from truth of SM_t, truth of SM_t from truth of EU_t, or truth of FR_t from truth of both SM_t and EU_t. This is a non-claim of implication and not a demonstration that the implications fail. To show that they fail, one would have to construct a system satisfying all assumptions and constitutive requirements of §2 and exhibit a time at which the antecedent is true and the consequent false; that construction is not carried out in the paper.